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1 字串

1.1 字典樹

```

1 struct trie{
2     trie *nxt[26]={};
3     int cnt=0;
4     int sz=0;
5 };
6 trie *root=new trie();
7 void insert(const string &s){
8     trie *now=root;
9     for(auto i:s){
10         if(now->nxt[i-'a']==NULL){
11             now->nxt[i-'a']=new trie();
12         }
13         now=now->nxt[i-'a'];
14     }
15 int qry(string &s){
16     trie *now=root;
17     for(auto i:s){
18         if(now->nxt[i-'a']==NULL) return;
19         now=now->nxt[i-'a'];
20     }
21     return now->cnt;
22 }
```

2 最短路徑

2.1 basic

```

1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }
```

3 圖論

3.1 凸包問題

```

1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return
5         (x<other.x) || (x==other.x && y<other.y);
6     };
7 };
8 struct ConvexHull{
9     vector<node> vec;
10    int n;
```

```

11 void init(vector<node> &v){
12     vec=v;
13     n=v.size();
14 }
15 int cross(node a,node o,node b){
16     return
17     (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
18 }
19 vector<int> point;
20 vector<int> area;
21 void build(){
22     point=vector<int>(n+1);
23     area=vector<int>(n+1);
24     vector<node> up,down;
25     int upsum=0;
26     int downsum=0;
27     for(int i=0;i<n;i++){
28         while(up.size()>=2 &&
29             cross(up[up.size()-2],
30                 up[up.size()-1],
31                 vec[i])<=0){
32             upsum-=cross(up[up.size()-2],
33                         {0,0},up[up.size()-1]);
34             up.pop_back();
35         }
36         up.push_back(vec[i]);
37         if(up.size()>1)
38             upsum+=cross(up[up.size()-2],
39                         {0,0},
40                         up[up.size()-1]);
41
42         while(down.size()>=2 &&
43             cross(down[down.size()-2],
44                 down[down.size()-1],
45                 vec[i])>=0){
46             downsum-=cross(down[down.size()-2],
47                           {0,0},down[down.size()-1]);
48             down.pop_back();
49         }
50         down.push_back(vec[i]);
51         if(down.size()>1)
52             downsum+=cross(down[down.size()-2],
53                           {0,0},down[down.size()-1]);
54
55         area[i+1]=abs(upsum-downsum);
56
57         if(i+1==1) point[i+1]=1;
58         else point[i+1]=up.size()+down.size()-2;
59     }
60 double getarea(int p){
61     return area[p]/2.0;
62 }
63 int getnum(int p){
64     return point[p];
65 }
66 };
```

4 數學

4.1 公式

• 中文測試

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$